

Hitec University Taxila, Department of Computer Engineering

Lab Manual for Mobile Application Development

Lab-11 Android Services

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1 Introduction

A Service is an application component that can perform long-running operations in the background. It does not provide a user interface. Once started, a service might continue running for some time, even after the user switches to another application. Additionally, a component can bind to a service to interact with it.

For example, a service can handle network transactions, play music, perform file I/O, or interact with a content provider, all from the background.

2 Activity Time boxing

Table 1: Activity Time Boxing

Task#	Activity Name	Activity time	Total Time
1	Introduction	60mins	60min
2	Walk Through Task	60mins	60min
3	Evaluation Task	60min	60min

3 Objectives of the experiment

- To get basic understanding regarding android services.
- To get basic understanding regarding how to start and stop android services.
- To get to know about the benefits related to android services.

4 Walk through Task

Creating New Activity

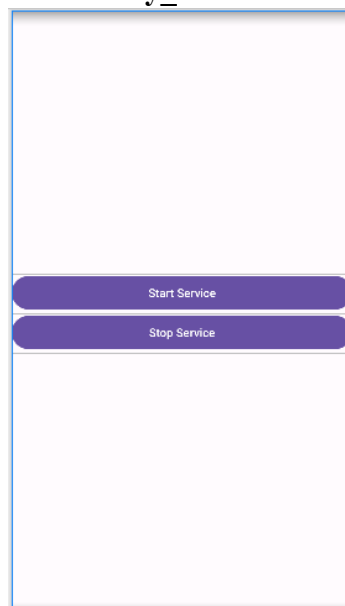
Step 1: Create Activities:

Create a new project just like you have created in your previous labs and create one more empty activity.

Step 2: Design XMLs of Activity:

Designs of activity are given below, just use the given xml code which is given in this step.

activity_main.xml



```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <Button
        android:id="@+id/start_service_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:onClick="startService"
        android:text="Start Service" />

    <Button
        android:id="@+id/stop_service_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:onClick="stopService"
        android:text="Stop Service" />
</LinearLayout>

```

Step 3: Write “MyService.java” class

```

import android.app.Service;
import android.content.Intent;
import android.os.IBinder;
import android.util.Log;

import androidx.annotation.Nullable;

public class MyService extends Service {
    @Nullable
    @Override
    public IBinder onBind(Intent intent) {
        return null;
    }

    @Override
    public int onStartCommand(Intent intent, int flags, int startId) {
        serviceThread thread = new serviceThread(startId);
        thread.start();
        return START_STICKY;
    }

    @Override
    public void onDestroy() {
        super.onDestroy();
        Log.d("MyService", "onStartCommand: Service Stopped");
    }

    class serviceThread extends Thread {
        int threadID;

        serviceThread(int threadID) {
            this.threadID = threadID;
        }
    }
}

```

```

        @Override
        public void run() {
            try {
                for (int i = 0; i < 10; i++) {
                    Log.d("MyService", "serviceRunnable Running i = " + i);
                    Thread.sleep(1000);
                }
            } catch (Exception e) {
                Log.d("MyService", "serviceRunnable Error " + e);
            }
            stopSelf(threadID);
        }
    }
}

```

Step 4: Write “MainActivityService.java” class

```

import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;

public class MainActivityService extends AppCompatActivity {

    Intent serviceIntent;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main_service);
    }

    public void startService(View view){
        serviceIntent = new Intent(getApplicationContext(), MyService.class);
        startService(serviceIntent);
    }

    public void stopService(View view){
        serviceIntent = new Intent(getApplicationContext(), MyService.class);
        stopService(serviceIntent);
    }
}

```

Step 3: Initialize “MyService” in AndroidManifest.xml

```
<service android:name=".MyService"/>
```

Step 6: Now Run Project

Click “Run” or press “Shift + F10”.